```
import pandas as np
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
df=pd.read_csv('/content/gender_submission.csv')
```

```
df.isnull().sum()#imform about null and missing values
```

```
from sklearn.preprocessing import LabelEncoder
encoder=LabelEncoder()
for col in df.select_dtypes(include='object').columns:
  data[col]=encoder.fit_transform(data[col])
```

```
x=df.iloc[:,0]
y=df.iloc[:,1]
```

```
import numpy as np
x=np.array(x).reshape(-1,1)
y=np.array(y).reshape(-1,1)
print("x array:",x)
print("y array:",y)
```

```
from sklearn.model_selection import train_test_split
```

```
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_state=42)
```

```
x_trainx_train
```

```
from sklearn.linear_model import LogisticRegression
```

```
model=LogisticRegression()
```

```
model.fit(x_train,y_train)
```

```
y=pred=model.predict(x_test)
```

```
from sklearn.metrics import accuracy_score,confusion_matrix,precision_score,recall_score,f1_score
```

```
accuracy =accuracy_score(y_test,pred)
print("Accuracy:",accuracy)
```

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
cm=confusion_matrix(y_test,y_pred,labels=[1,0])
```

```
plt.figure(figsize=(6,4))
sns.heatmap(cm,annot=True,fmt='d',cmap='rainbow',xticklabels=['predicted 1','predicted'],yticklabels=['actual 1','actual 0'])
plt.xlabel('predicted label')
plt.ylabel('actual label')
plt.title('confusion matrix heatmap')
plt.show()
```

```
from sklearn.metrics import roc_curve,auc
```

```
y_prob=model.predict(x_test)[:,1]
```

Start coding or generate with AI.